

No. of Functional Features Included in the Solution

Date	30 June 2026
Team ID	XXXXXX
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Home Page	Displays project overview and navigation	UI	Done	High
2	Flood Prediction	Predicts flood probability using XGBoost	ML Model	Done	High
3	Risk Analysis	Classifies prediction into Low, Moderate or High Risk	Backend	Done	High
4	Safety Recommendations	Displays precautions based on prediction	Backend	Done	Medium
5	About Page	Provides project and model information	UI	Done	Medium
6	Responsive Web Design	Supports desktop and mobile devices	Frontend	Done	Medium
7	Model Integration	Connects trained XGBoost model with Flask	Backend	Done	High
8	Result Dashboard	Displays probability, risk level and recommendations	UI	Done	High

Feature Summary:

Metric	Count/Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	Flood Prediction, Risk Analysis, Model Integration, Result Dashboard
Additional / Nice-to-Have Features	Responsive UI, Professional Dashboard, About Page, Safety Recommendations
Features Tested & Verified	All implemented features tested successfully.

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	4	Home, Predict, Result, About Pages
2	Backend / Logic	3	Prediction Logic, Risk Analysis, Recommendations
3	Database / Storage	1	Trained XGBoost Model (.pkl)
4	API / Integration	1	Flask Route Integration
5	Security / Authentication	0	Not Applicable